Payman Tohidifar

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Summary

Computational biologist and biomolecular engineer with 6+ post-graduate years of experience spanning computational science, AI/ML, and experimental wet-lab molecular biology applied to protein engineering and small-molecule biosynthesis. Hands-on experience includes end-to-end ML pipelines for sequence-based protein function/fitness prediction, LLM architectures built and trained from scratch, orchestration of state-of-the-art generative AI models for de novo protein design, automated processing and statistical analysis of large biological datasets, and software development in professional environments. Co-inventor on 6 filed patent applications and author of 8 peer-reviewed publications. An independent researcher and proactive learner, with a passion for building and applying AI to automate and accelerate the biological discovery pipeline for real-world impact. Experience leading cross-functional projects, with a track record of establishing new capabilities and scientific discoveries in fast-moving, high-adaptability environments.

Technical Skills

Programming & ML Frameworks: Python, R, Bash, C++; PyTorch, JAX/Flax, TensorFlow/Keras (familiar), Weights & Biases (familiar)

Data Science & SWE Tools: NumPy, SciPy, Pandas, Scikit-learn; CI/CD, pytest, Git, Docker, Snakemake, Pixi/Conda, Claude Code, HPC/Slurm

Structural Biology & Bioinformatics: BioPython, PyMOL, HMMER, BLAST, MMseqs2, Foldseek, Common Biomolecules Databases

Wet-lab: Protein engineering/expression, directed evolution, HTP enzyme assays, flow cytometry/FACS, CRISPR editing

Experience

Scientist Feb 2022 – Present
BP (Biosciences, Technology and Innovation) *San Diego, CA*

- Engineered an end-to-end, ML pipeline for optimization of industrial enzymes in low-N regimes, combining unsupervised VAE and protein language model (pLM) fitness representations for zero-shot variant scoring with supervised models trained on experimental activity and thermostability measurements.
- Applied the pipeline within DBTL cycles alongside bench scientists to prioritize variants for experimental testing; extending it into an iterative active-learning loop (in development).
- Co-led development of a semi-automated HTP directed-evolution and eukaryotic protein-expression pipeline, cutting turnaround time from 6 weeks to 2 weeks and accelerating the DBTL cycle org-wide.
- Architected a full-stack platform for ingestion, consolidation, statistical analytics, and visualization of multi-modal biological data across projects; selected as a top-3 platform for executive leadership review (*Link to GitHub Repo for a prototype*).
- Designed ML/statistical pipelines to analyze complex, multi-dimensional biorefinery data, directly informing key business decisions.
- Pioneered an HTP platform (a novel FACS approach) and led a team to boost protein production in an industrial eukaryotic host, identifying multiple enzyme hyperproducers for scaled testing and cutting discovery time from 1+ year to 3 months.
- Standardized and drove adoption of in-house Oxford Nanopore long-read sequencing and de novo genome assembly SOPs (EPI2ME, Nextflow)
- Led a 7-person cross-functional project on biosynthesis of value-added small molecules, collaborating across computational and experimental disciplines, and directly contributed to 6 patent applications within 2 years.
- Engineered a CRISPR-MAD7 platform for single-nucleotide-resolution genetic engineering, enabling construction of 50+ bacterial cell lines and replacing a technology costing over $10K/yr$.

Postdoctoral Research Associate Aug 2019 – Feb 2022
Institute for Genomic Biology, UIUC *Urbana, IL*

- Closed evolved Salmonella enterica strain's genome and called genetic variants via whole-genome sequencing (WGS) analysis using GATK, mapping the genetic drivers of NAC resistance and virulence attenuation and contributing to a publication.
- Co-developed a modular genetic toolkit for multi-gene expression and CRISPR-Cas9 chromosomal editing in two non-model yeasts, enabling biosynthesis of value-added small molecules (2 publications, 1 conference paper).

Graduate Research Assistant Jan 2012 – May 2019
Department of Chemical & Biomolecular Engineering, UIUC *Urbana, IL*

• Conceptualized and led three independent projects elucidating molecular mechanisms of unconventional chemoreceptor sensing in bacteria — bidirectional pH, alcohol, and toxic aromatic molecule sensing — combining protein engineering and mutagenesis with structural biology, MD simulations of receptor conformational dynamics, and bioinformatics; resulted in 3 journal publications and multiple conference presentations.

• Led discovery and characterization of an unconventional DNA-sensing receptor in *B. subtilis* using in vivo/in vitro assays and in silico motif-discovery analysis; awarded Best Presentation. (*Link to project details*)

• Co-developed a modular C++ framework for multiscale simulation of cellular population behavior in response to complex spatiotemporal chemical gradients (*Link to project details*).

Independent LLM & Protein ML Projects

Containerized De Novo Protein Binder Design Pipeline | *Link to GitHub Repo*

• Architected a generative design pipeline orchestrating diffusion-based backbone generation (RFdiffusion), inverse-folding sequence design (ProteinMPNN), dual-engine monomer self-consistency validation (ESMFold, Boltz-2), and co-folding interface scoring (Boltz-2, AlphaFold2-Multimer) into a tiered, confidence-score-gated binder-design funnel; prototyped against the human insulin-like receptor E1 domain targeting three defined hotspots.

• Unified heterogeneous tool outputs into a single metrics table, automating extraction of the confidence scores that drive each selection gate; extending the platform to Snakemake for automated, production-scale execution.

Deep Learning for Protein Function Prediction | *Link to GitHub Repo*

• Built a modular deep learning platform (PyTorch, JAX/Flax) on the CAFA benchmark to predict multi-label protein function across 291 Gene Ontology terms from sequence alone, integrating ESM-2 transformer embeddings into an extensible framework for rapid experimentation with custom architectures, loss functions, and evaluation metrics.

Production-Ready GPT-2 Text Classifier From Scratch | *Link to GitHub Repo*

• Implemented the GPT-2 architecture from scratch in PyTorch with a configuration-driven training pipeline, LoRA parameter-efficient fine-tuning, multi-GPU scaling via Distributed Data Parallel (DDP), Weights & Biases tracking, and an end-to-end evaluation suite.

Lung Cancer Bulk RNA-Seq Data Analysis Pipeline | *Link to GitHub Repo*

• Built an end-to-end Python/R pipeline for CCLE bulk RNA-seq data — count processing, PCA, hierarchical clustering, differential expression, and pathway enrichment.

Publications

Co-first authors are denoted with asterisks.

8. Koh, HG.*, **Tohidifar, P.***, Oh, H., Ye, Q., Jung, SC., Rao, CV., Jin, YS. (2025) RT-EZ: A Golden Gate Assembly Toolkit for Streamlined Genetic Engineering of *Rhodotorula toruloides*. ACS Synth. Biol.

7. Hamed, S., Emara, M., **Tohidifar, P.**, Rao, CV. (2025) N-Acetyl Cysteine Exhibits Antimicrobial and Anti-Virulence Activity Against *Salmonella enterica*. PLoS One.

6. Yook, S., Deewan, A., Ziolkowski, L., Lane, S., **Tohidifar, P.**, Cheng, M., Singh, V., Stasiewicz, MJ., Rao, CV., Jin, YS. (2025) Engineering and Evolution of *Yarrowia lipolytica* for Producing Lipids from Lignocellulosic Hydrolysates. Bioresource Technology.

5. Bodhankar, GA., **Tohidifar, P.**, Foust, ZL., Ordal, GW., Rao, CV. (2022) Characterization of Opposing Responses to Phenol by *Bacillus subtilis* Chemoreceptors. J Bacteriol.

4. Joseph, E.*, **Tohidifar, P.***, Sarver, CT., Mackie, RI., Rao, CV. (2022) Fundamentals of Polymer Biodegradation Mechanisms. Wiley (Book chapter).

3. **Tohidifar, P.**, Bodhankar, GA.*, Pei, S.*, Cassidy, CK., Walukiewicz, HE., Ordal, GW., Stansfeld, PJ., Rao, CV. (2020) The Unconventional Cytoplasmic Sensing Mechanism for Ethanol Chemotaxis in *Bacillus subtilis*. mBio.

2. **Tohidifar, P.**, Plutz, MJ., Ordal, GW., Rao, CV. (2020) The Mechanism of Bidirectional pH Taxis in *Bacillus subtilis*. J Bacteriol.

1. Walukiewicz, HE., **Tohidifar, P.**, Ordal, GW., Rao, CV. (2014) Three Adaptation Systems of *B. subtilis* Chemotaxis. Molecular Microbiology.

Patents

4. "Recombinant microorganisms for beta-myrecene and nerol production." Valle, F., Vu, HNN., **Tohidifar, P.** Patent Application Pub. No. WO/2026/122843 A1, published June 11, 2026.

3. "Engineered dxp pathway for improved isoprenoid production in *E. coli*." Valle, F., **Tohidifar, P.**, Najdi, T., Vu, HNN. Patent Application Pub. No. WO/2025/155822 A1, published July 24, 2025.

2. "Recombinant microorganisms with increased accumulation and/or flux of cytidine triphosphate (CTP)." Valle, F., **Tohidifar, P.**, Najdi, T. Patent Application Pub. No. WO/2025/111429 A1, published May 30, 2025.

1. "Gluconate-regulated expression systems." Valle, F., **Tohidifar, P.** Patent Application Pub. No. WO/2025/059030 A1, published March 20, 2025.

• Co-invented 2 provisional patent applications with colleagues, covering novel enzymes and metabolic pathways for the biosynthesis of value-added small molecules (filed Feb 2 and Feb 19, 2026).

Education

University of Illinois at Urbana-Champaign (UIUC) Urbana, IL
Ph.D. in Chemical & Biomolecular Engineering *2019*
- Computational Science & Engineering Certificate, 2019
- Dissertation: *Non-canonical sensing mechanisms in bacteria*

Sharif University of Technology Tehran, Iran
B.S. in Chemical Engineering *2011*

Coursework & Certifications

• Self-Study: *Deep Learning for Biology*, C. Ravarani and N. Latysheva (O'Reilly Media, 2025) — in progress (*Notes*)

• *Claude Code: A Highly Agentic Coding Assistant*, DeepLearning.AI (2026)

• Deep Learning Specialization, Coursera (2026) (*Credential*, *Notes*)

• Self-Study: *Build a Large Language Model (From Scratch)*, S. Raschka (Manning Publications, 2024) — completed 2026 (*Notes*)

• Machine Learning Specialization, Coursera (2025) (*Credential*, *Notes*)

• Mathematics for Machine Learning and Data Science Specialization, Coursera (2025) (*Credential*)

• bp Data Science Intermediate Bootcamp, Pragmatic Institute (2024) (*Credential*)

• Computational Science & Engineering Certificate, UIUC (2019) — Numerical Analysis (CS450), Numerical Methods for PDEs (CS555), Introduction to Optimization (CSE441), Nonlinear Dynamics (PHYS510)

• ChIP-Seq Analysis Hands-On Training, High-Performance Biological Computing, UIUC (2017)

Selected Conference Talks

• **Tohidifar, P.**, Kimura, Y., Rao, CV. (2015) A modular computational framework for multi-scale simulation of chemotaxis in complex environments. AIChE, Salt Lake City, UT.

• **Tohidifar, P.**, Walukiewicz, HE., Ordal, GW., Rao, CV. (2016) DNA taxis in *Bacillus subtilis*. Receptor Fest 19th Annual Meeting, Santa Barbara, CA.

• **Tohidifar, P.**, Walukiewicz, HE., Ordal, GW., Rao, CV. (2017) DNA sensing in *Bacillus subtilis*. BLAST XIV, New Orleans, LA.

Honors & Awards

• Nucleic Acids Research Award for an Outstanding Poster/Oral Presentation by a Young Investigator, BLAST XIV, New Orleans, LA, 2017.

• Graduate College Conference Travel Award, UIUC, 2016.

• Graduated with honors (ranked 2nd/120), Department of Chemical Engineering, Sharif University of Technology, Iran, 2011.

• Singapore International Pre-Graduate Award, ICES, A*STAR, Singapore, 2010.

Professional Service

Reviewer for PLoS Computational Biology and PLoS One, 2018 – present.